

Section: Particles and Waves

Topic: Radiation, The Inverse Square Law

A radioactive source emits gamma radiation. The radiation is detected using a GM (Geiger-Müller) tube connected to a counter.

(a)

Explain why the count rate measured by the detector must be corrected for background radiation

✓ The GM tube detects **all ionising radiation**, including background radiation from natural sources (like rocks, cosmic rays). To get an accurate measurement of the radiation from the **source only**, we subtract the background count rate from the total.

(b)

The corrected count rate is measured at various distances from the source.

What is the expected relationship between count rate C and distance d ?

✓ The gamma radiation follows the **inverse square law**:

$$C \propto \frac{1}{d^2}$$

This means: if distance doubles, the count rate becomes one-quarter.

(c)

At a distance of 0.25 m, the corrected count rate is 1600 counts per minute.

Calculate the count rate at a distance of 0.50 m

Using the inverse square law:

$$\frac{C_1}{C_2} = \left(\frac{d_2}{d_1}\right)^2 = \left(\frac{0.50}{0.25}\right)^2 = 4 \Rightarrow C_2 = \frac{C_1}{4} = \frac{1600}{4} = 400 \text{ counts/min}$$

 Revision Tips

- Always correct radiation readings for **background** before analysis
- Gamma radiation behaves like light — its intensity decreases with distance squared
- The **inverse square law** is:

$$I \propto \frac{1}{r^2}$$

- To compare intensities or count rates at different distances, set up a ratio:

$$\frac{C_1}{C_2} = \left(\frac{d_2}{d_1}\right)^2$$